

MANGU 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

232/1

PHYSICS

PAPER 1 (THEORY)

TIME: 2 HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

1. Write your **name, admission number and index number** in the spaces provided
2. Sign and write the date of examination in the spaces provided
3. This paper consists of **TWO** sections **A** and **B**
4. Answer **ALL** questions in the spaces provided
5. Answers must be in ink and English
5. Mathematical table and electronic calculators may be used.
6. **ALL** working **MUST** be shown clearly where necessary.

Take: acceleration due to gravity $g = 10\text{m/s}^2$

FOR OFFICIAL USE ONLY

SECTION	QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORES
A	1 - 12	25	
B	13	11	
	14	11	
	15	13	
	16	09	
	17	11	
	TOTAL SCORE	80	

For Marking Schemes Contact 0746 222 000 / 0742 999 000

SECTION A (25 MARKS)

Attempt ALL questions in this section.

1. Name the branch of physics that deals with solar water heating system.

(1 mark)

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2. Define the term “Ideal gas” as used in gas laws

(1 mark)

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3. At Thompson falls water, falls through a height of 30m. Calculate the final temperature if the temperature at the top is 14°C (Take $g = 10 \text{ N/kg}$ and s.h.c of water = 4200 J/kgK) **(3 marks)**

4. State two conditions that must be fulfilled for a body to be in **equilibrium**

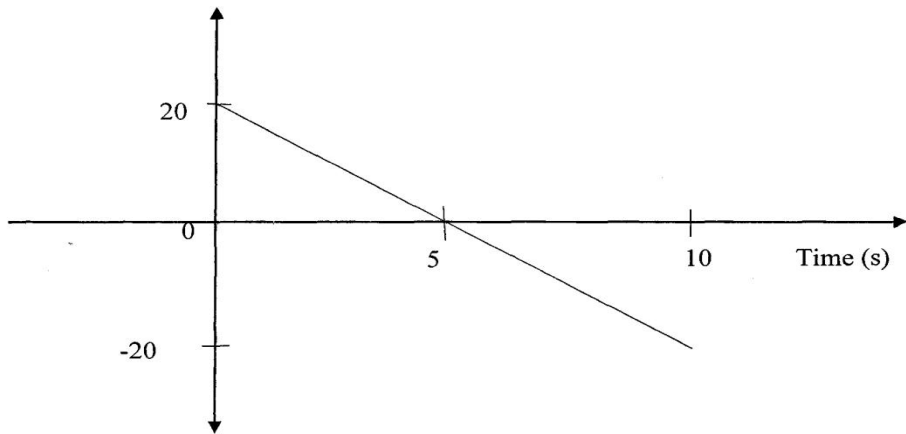
(2 marks)

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5. The figure below shows a graph of velocity against time for a moving



body.

Determine the distance covered by the body during the 10 seconds (3 marks)

6. Smoke is enclosed in smoke cell and sealed. When illuminated and viewed under a microscope, it is observed to be moving in continuous random motion. State and explain the observation when cold water is poured on the smoke cell. (2 marks)

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7. The pressure inside a submarine is maintained at 500m of mercury(500mHg). Determine the depth at which the submarine will be safe from implosion taking the density of water as 1.025g/cm³ and atmospheric pressure = 760cmHg. (Take the density of mercury as 13600 Kg/m³) (3 marks)

8. Explain why two cars are likely to crush as they travel at high velocities sideways. **(2 marks)**

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9. When a body of mass 0.25kg is acted on by a force, its velocity changes from 5.0 m/s to 7.5m/s, determine the impulse of the force. **(2 marks)**

10. State two factors that affect the spring constant of springs made from the same wire. **(2mks)**

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11. Draw a well labeled diagram of a clinical thermometer. **(3 marks)**

12. Differentiate between solids, liquids and gases in terms of thermal conductivity. (1 mark)

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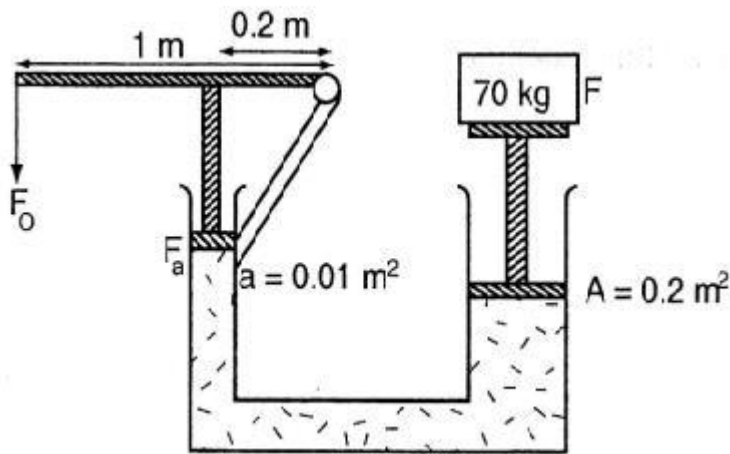
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SECTION B (55 MARKS) Attempt ALL questions in this section.

13. The figure below shows a hydraulic press supporting a load F.



a) What properties of oils make them suitable for use in hydraulic machines such as the one above? (2marks)

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b) If A and a are areas of cross-section of the pistons, and the lengths of the arm are as given, find: i.

The force F_a
(2 marks)

ii. The force F_0
(2 marks)

c) The mechanical advantage of the machine
(1mark)

d) The velocity ratio of the machine
(3 marks)

e) State one reason why the efficiency of a pulley system is always less than 100% (1 mark)
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14. a) State Newton's third law of motion. (1 mark)

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b) Distinguish between elastic and inelastic collision. (2 marks)

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c) A mini bus of mass 2000kg travelling at a constant velocity of 20m/s collides with a stationary car of mass 1000kg. The impact takes 2 seconds before the two moves together at a constant velocity for half a minute. Calculate.

i) The common velocity (3 marks)

ii) The distance moved after impact. (2 marks)

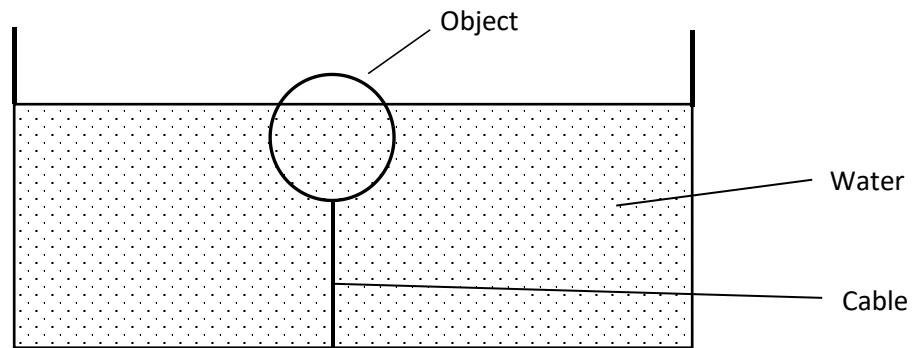
iii) The change in Kinetic energy. (3 marks)

15. a) State the law of floatation.

(1 mark)

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b) The figure below shows a floating object of volume 4000cm^3 and mass 10g . It is held as shown in water of density 1.25g/cm^3 by a light cable at the bottom so that $\frac{3}{4}$ of the volume of the object is below the water surface (Assume the upthrust due to air is negligible).



Calculate the volume of the object under water.

(2 marks)

ii) Calculate the weight of water displaced.

(3 marks)

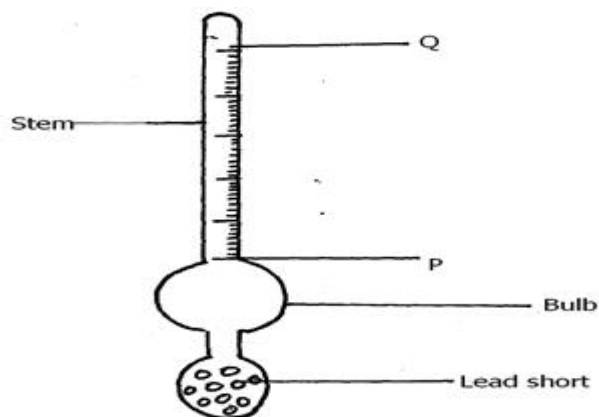
iii) Determine the tension in the cable.

(2 marks)

iv) Calculate the density of the object.

(2 marks)

c) The diagram below shows a car acid hydrometer.



(i) Indicate on the diagram above the minimum and the maximum measurement to be taken.

(1 mark)

(ii) State the reason why

I. The bulb is wide.

(1 mark)

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II. Lead shots are used

(1 mark)

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16. (a) State what is meant by the term 'specific latent heat of vaporization'

(1mark)

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b) In an experiment to determine the specific latent heat of vaporization of water, steam at 100°C was passed into water at 5°C contained in a well-lagged copper calorimeter. The following measurements were made;

Mass of calorimeter 50g

Initial mass of water 70g

Final mass of calorimeter + water + condensed steam = 123g

Final temperature of mixture = 30°C

(Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$, specific heat capacity for copper = $390 \text{ J kg}^{-1} \text{ K}^{-1}$)

(i) Determine the:

I. Mass of condensed steam

(2 marks)

II. Heat gained by the calorimeter and water

(3 marks)

(ii) Given that L is the specific latent heat of vaporization of steam, determine the value of L .

(3 marks)

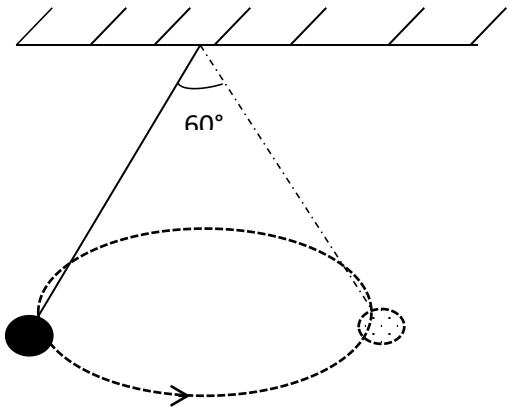
17. a) Define the term angular velocity.

(1 mark)

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b) A body of mass 5 kg is whirled horizontally making a conical pendulum as shown in the diagram below



i. State what happens when the speed of whirling is increased (1 mark)

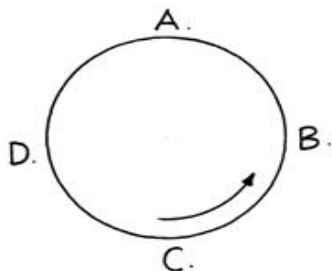
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ii. If the radius of the path is 50cm and the centripetal force is 285N. Determine the linear velocity of the mass

(3 marks)

- c) A stone is whirled in a vertical circle as shown in the figure below using a string of length 40 cm. A, B, C and D are various positions of the stone in its motion. The stone makes 2 revolutions per second and has a mass of 100g.



i) Calculate:

I. The angular velocity

(2 marks)

II. The tension on the string at position A

(3 marks)

- (ii) At C where the stone has acquired a constant angular speed, the string cuts. What is the name given to the path followed by the stone?

(1 mark)

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